

Docker:

A lot changed in a year

Chris Swan, CTO

@cpswan

cohesiveFT

Cloud native networking

TL;DR

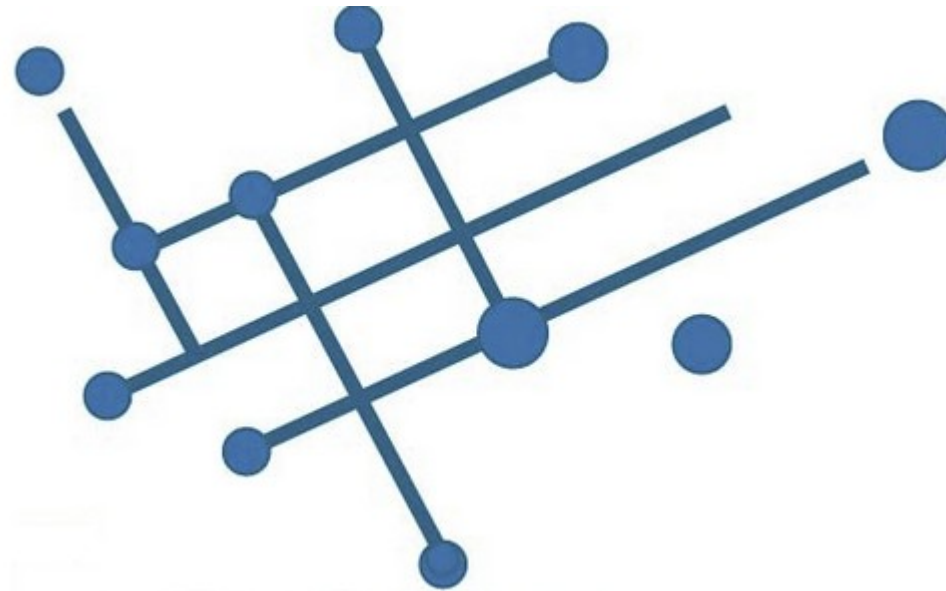
Docker is important for DevOps because it tightens iteration loops

The core project developed very fast, but the ecosystem is now moving faster

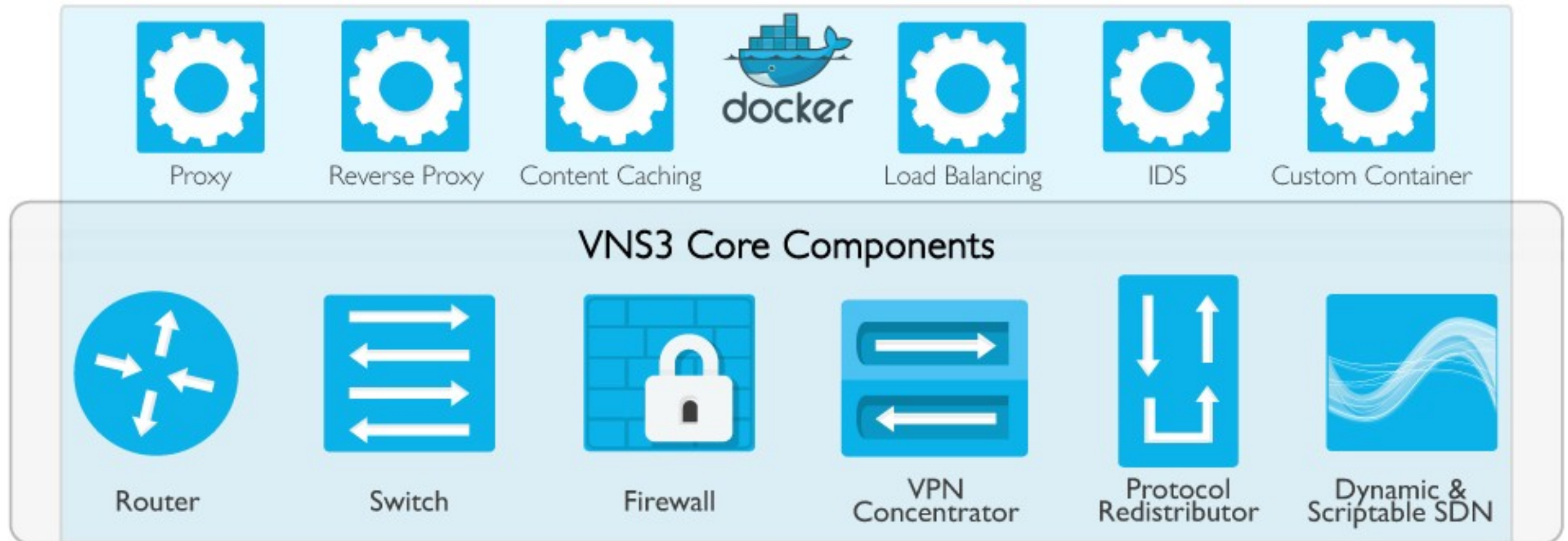
They did it with Docker

Why me?

Grid, Cloud, Docker

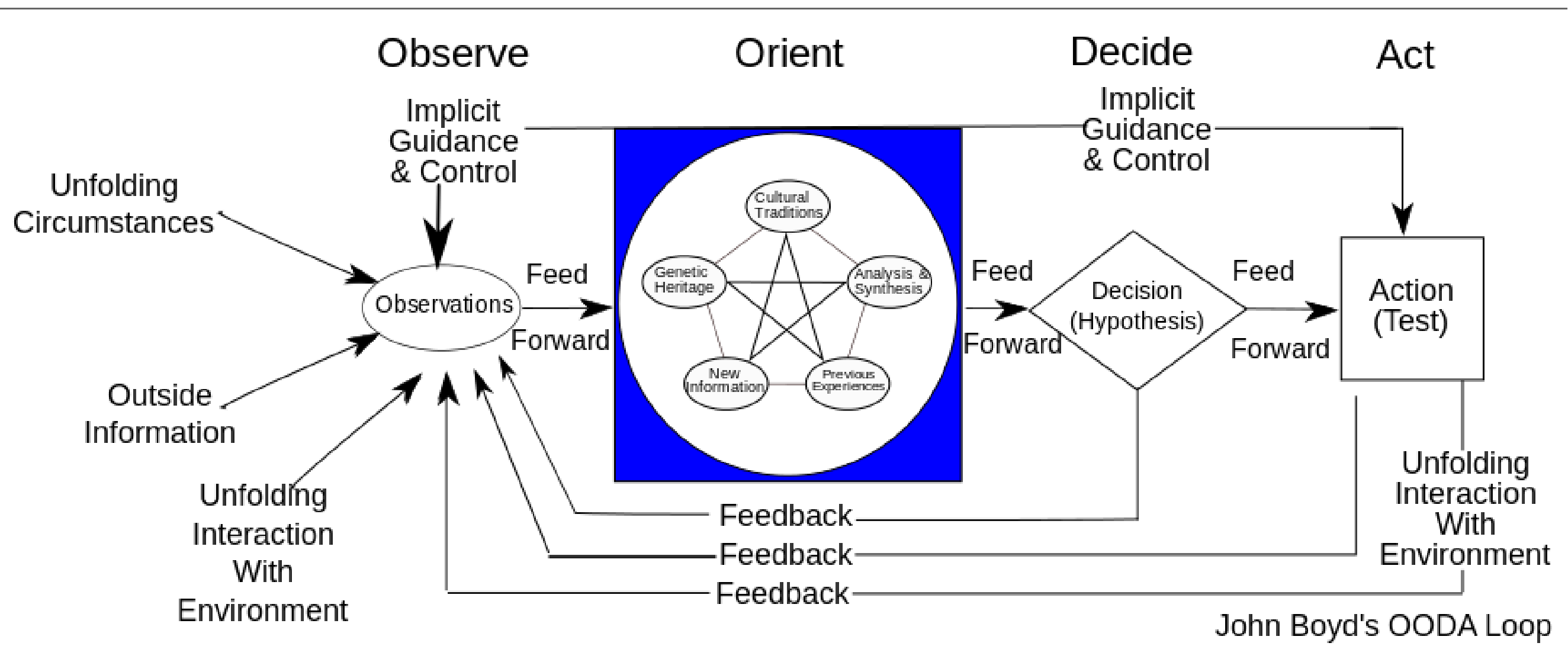


Docker inside



This is a DevOps track, so
why do we care about Docker?

John Boyd's OODA loop



In the beginning

0.1.0 (2013-03-23)

Used LXC

Registry to push/pull images

Where were we a year ago?

0.6.6 (2013-11-06)

Runtime

- Ensure container name on register
- Fix regression in /etc/hosts
- Add lock around write operations in graph
- Check if port is valid
- Fix restart runtime error with ghost container networking
- Add some more colors and animals to increase the pool of generated names
- Fix issues in docker inspect
- Escape apparmor confinement
- Set environment variables using a file.
- Prevent docker insert to erase something
- Prevent DNS server conflicts in CreateBridgelface
- Validate bind mounts on the server side
- Use parent image config in docker build
- Fix regression in /etc/hosts

Client

- Add -P flag to publish all exposed ports
- Add -notrunc and -q flags to docker history
- Fix docker commit, tag and import usage
- Add stars, trusted builds and library flags in docker search
- Fix docker logs with tty

RemoteAPI

- Make /events API send headers immediately
- Do not split last column docker top
- Add size to history

Other

- Contrib: Desktop integration. Firefox usecase.
- Dockerfile: bump to go1.2rc3

We still didn't have

- A regular release cycle
- Compatibility with standard Linux distros
- Links
- Naming

Much of that came in 0.7.0 towards the end of Nov 2013

The watershed release

0.9.0 (2014-03-10)

Libcontainer

(LXC gets thrown under the bus)

0.9.1 (2014-03-24)

Ubuntu 14.04

Centos 7

Etc.

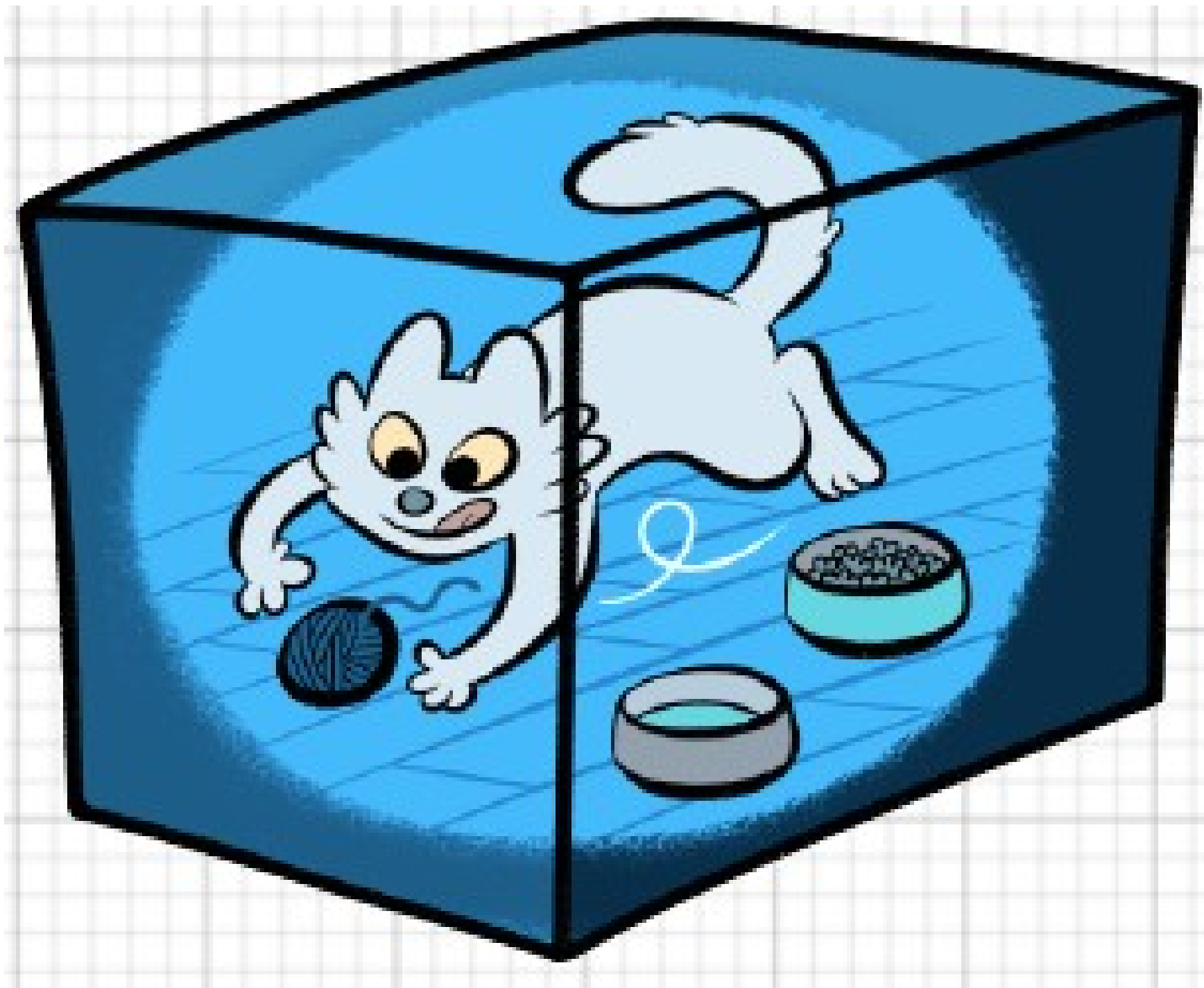
And then came DockerCon

1.0.0 (2014-06-09)

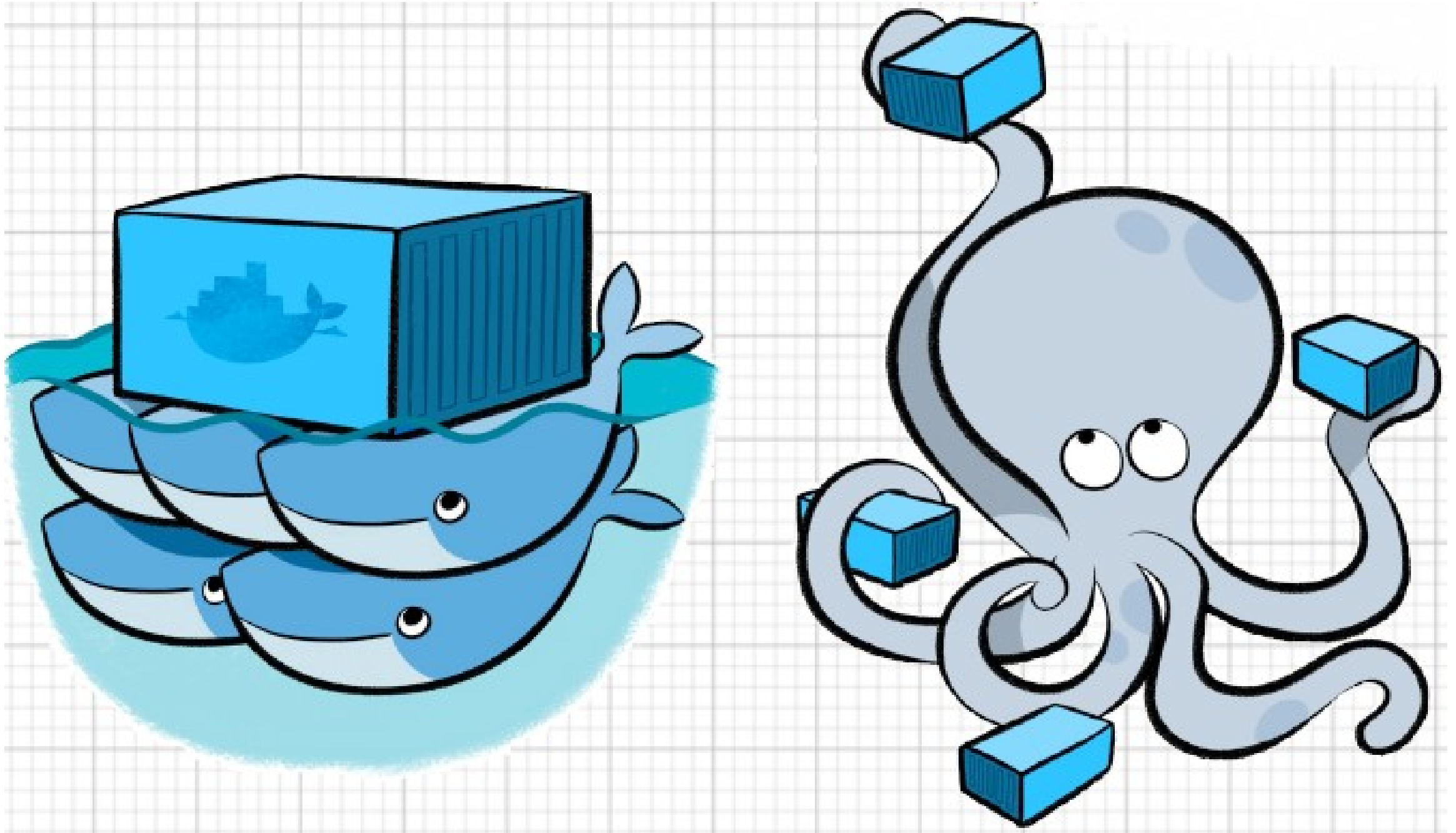
Notable features since 0.12.0 (2014-06-05)

- Production support

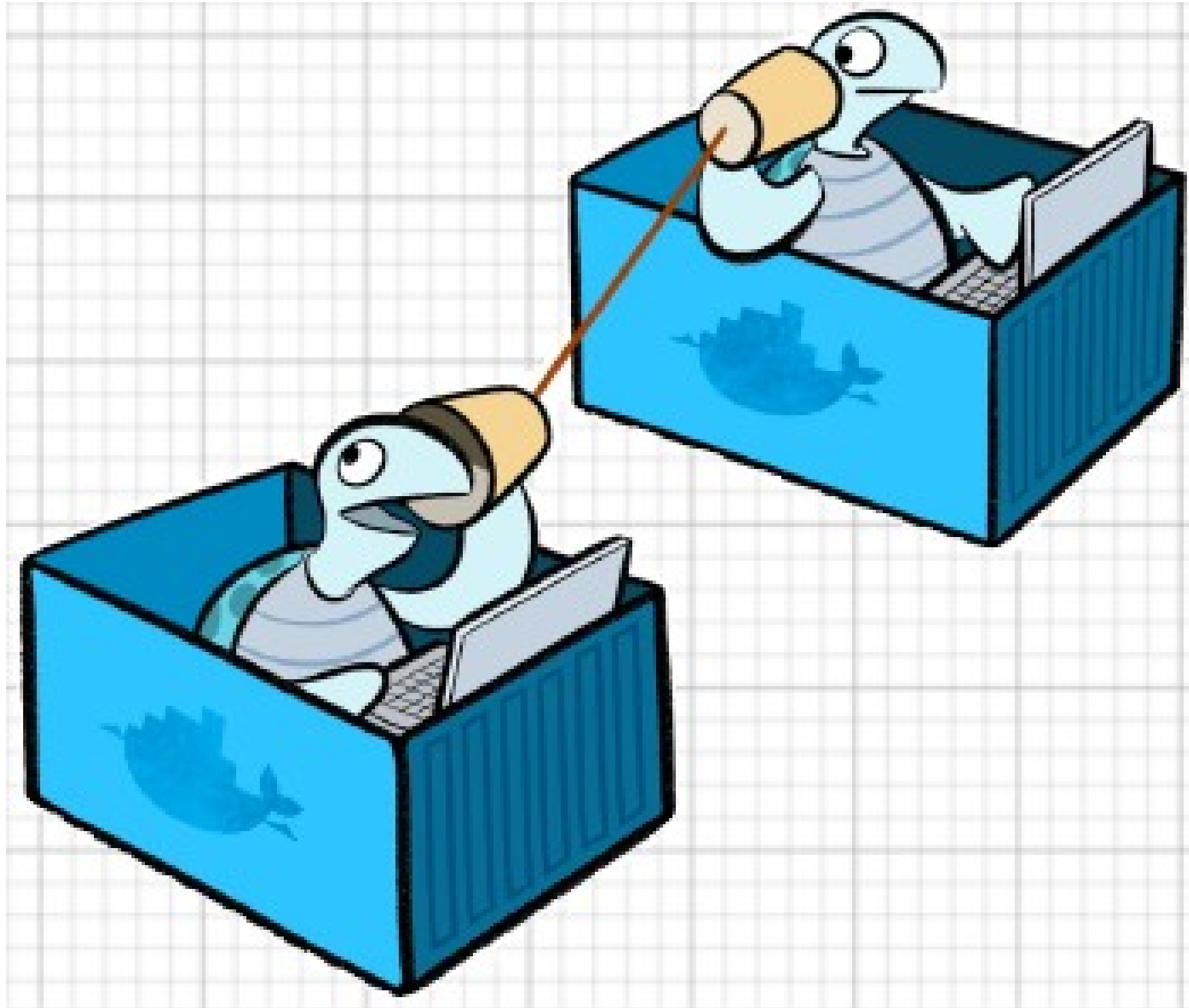
libcontainer



libswarm



libchan



Since 1.0.0

exec

create

Shocker, Flocker, Clocker

Shocker

The issue

```
/* shocker: docker PoC VMM-container breakout (C) 2014 Sebastian Krahmer
 *
 * Demonstrates that any given docker image someone is asking
 * you to run in your docker setup can access ANY file on your host,
 * e.g. dumping hosts /etc/shadow or other sensitive info, compromising
 * security of the host and any other docker VM's on it.
```

<http://stealth.openwall.net/xSports/shocker.c>

The response

 **Hacker News** new | threads | comments | show | ask | cpswan (426) | logout
jobs | submit

shykes 27 days ago | link | parent | flag

Hi all, I'm a maintainer of Docker. As others already indicated this doesn't work on 1.0. But *it could have*.

Please remember that at this time, we don't claim Docker out-of-the-box is suitable for containing untrusted programs with root privileges. So if you're thinking "pfew, good thing we upgraded to 1.0 or we were toast", you need to change your underlying configuration now. Add apparmor or selinux containment, map trust groups to separate machines, or ideally don't grant root access to the application.

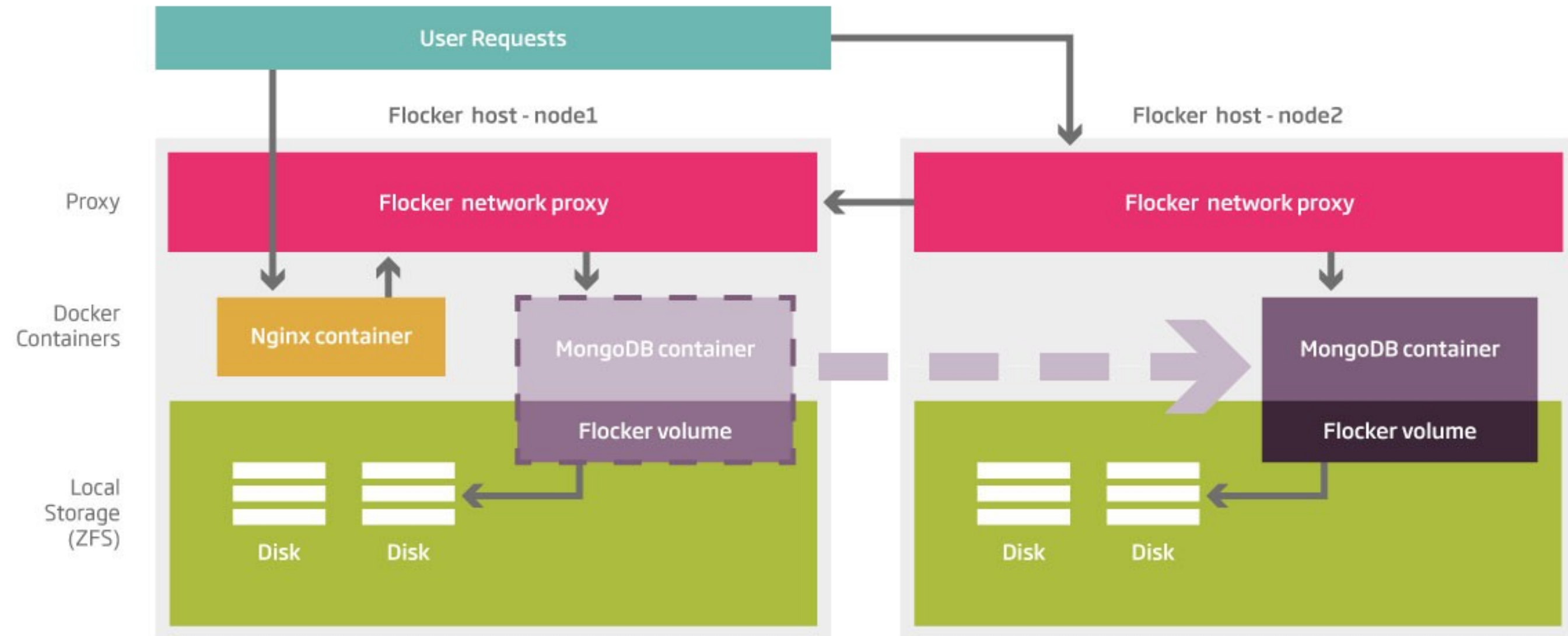
Docker will soon support user namespaces, which is a great additional security layer but also not a silver bullet!

When we feel comfortable saying that Docker out-of-the-box can safely contain untrusted uid0 programs, we will say so clearly.

<https://news.ycombinator.com/item?id=7910117>

Flocker

Flocker moves a database from node1 to node2, re-routes network connections to new location



Clocker

Create Application

X

Choose or build the application to deploy

Catalog

YAML

Other ▾

**Docker Cloud**
Deploys a Docker cloud infrastructure.

**JBoss**
Single JBoss server.

**ActiveMQ Broker**
Single ActiveMQ message broker

**Tomcat**
Single Tomcat server.

**Elastic Web Application**
Deploys a WAR to an Nginx load-balanced Tomcat cluster, with an

**Docker Cluster**
Simple Docker application cluster.

Next

copyright 2014

25

Composition

Fig



fig.yml

```
1  todomvcdb:
2    image: cpswan/todomvc.mysql
3    expose:
4      - "3306"
5    volumes:
6      - /data/mysql:/var/lib/mysql
7  todomvcapp:
8    image: cpswan/todomvc.sinatra
9    expose:
10     - "4567"
11    links:
12     - todomvcdb:db
13  todomvcssl:
14    image: cpswan/todomvc.ssl
15    ports:
16     - "443:443"
17    links:
18     - todomvcapp:app
```

geard



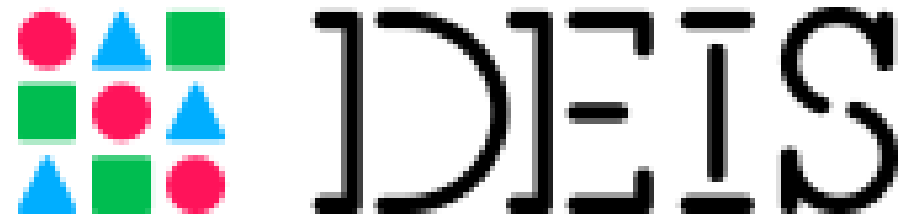
Panamax



```
CenturyLink Labs - http://www.centurylinklabs.com/
Please select one of the following options:
 1) init:          First time installing Panamax! - Downloads CoreOS VM and inst
alls latest Panamax version.
 2) pause:         Stops Panamax
 3) up:            Starts Panamax
 4) restart:       Stops and Starts Panamax.
 5) reinstall:     Deletes your applications and CoreOS VM; reinstalls to latest
Panamax version.
 6) info:          Displays version of your local panamax install.
 7) check:         Checks for available updates for Panamax.
 8) download:      Updates to latest Panamax version.
 9) delete:        Uninstalls Panamax, deletes applications and CoreOS VM.
10) ssh:           Ssh into Panamax CoreOS VM.
11) help:          Show this help
12) quit
#?
```

PaaS caught Docker

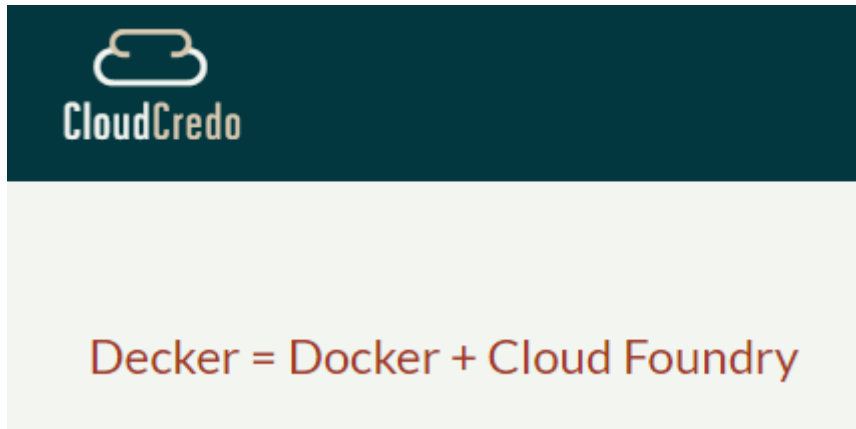
Docker based PaaS



Docker powered mini-Heroku in around 100 lines of Bash



Docker in PaaS



APCERA®

CONTINUUM

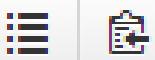


PaaS == Opinionated?

Docker might be the anti PaaS

branch: master

[dockerfiles](#) / [meetup](#) / **Dockerfile**



Chris Swan 2 minutes ago Dockerfile example for Chicago Docker Meetup

0 contributors

file | 18 lines (17 sloc) | 0.513 kb



Open

Edit

Raw

Blame

History

Delete

```
1 FROM ubuntu:12.04
2
3 MAINTAINER cpswan
4 # Add universe repository to /etc/apt/sources.list
5 # we need it for nginx
6 RUN sed -i s/main/'main universe'/ /etc/apt/sources.list
7 # Update repos so that changes can take effect
8 RUN apt-get update
9 # Install nginx
10 RUN apt-get install -y nginx
11 # Turn off daemon mode
12 RUN echo "\ndaemon off;" >> /etc/nginx/nginx.conf
13 # Meetup customisation
14 RUN sed -i s/nginx/'Chicago Docker Meetup'/ /usr/share/nginx/www/index.html
15 # Expose web server
16 EXPOSE 80
17 # Run nginx
18 CMD /usr/sbin/nginx
```

PaaS == Operationalised?

Orchestration

Everybody has one...



Brad Rydzewski

@bradrydzewski

what I learned at [#dockercon](#): everyone is building their own orchestration platform. seriously. everyone



Kubernetes is the one to watch



SOFTLAYER[®]
an IBM Company



Google Cloud Platform

This week...



Google Container Engine (Alpha)

Run Docker containers on Google Cloud Platform, powered by Kubernetes. Container Engine takes care of provisioning and maintaining the underlying virtual machine cluster, scaling your application, and operational logistics like logging, monitoring, and health management.

... not containerised cloud (yet)

Pricing

Container Engine uses Google Compute Engine instances for nodes in the cluster, including the cluster master (created automatically for every cluster) and the number of worker nodes you specify. You will be billed for those instances according to [Compute Engine's pricing](#), until the clusters are deleted. During this alpha phase, there is no additional charge for Container Engine on top of the Compute Engine pricing.

So why does it matter?

Build, Ship > Run?



Image credit <http://www.mediaagility.com/2014/docker-the-next-big-thing-on-cloud/>

Docker Hub

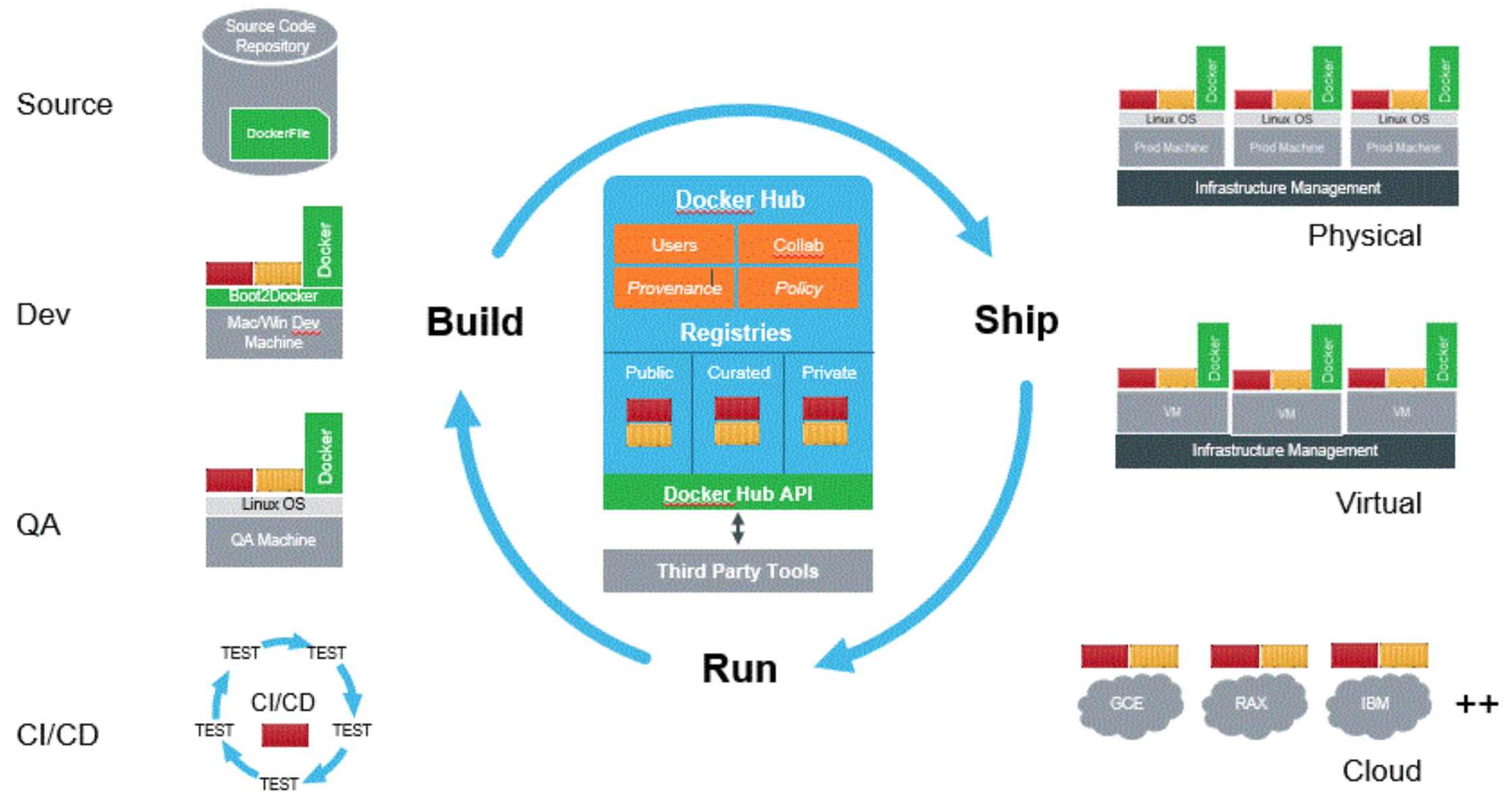


Image credit <http://blog.docker.com/2014/06/announcing-docker-hub-and-official-repositories/>

Bottled software



Image credit CC Julio Morales <https://www.flickr.com/photos/phooleo/4703216251>

There's still a bunch of work to be done

Containers don't contain



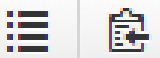
With thanks to Dan Walsh @rhatdan
Watch his DockerCon 2014 presentation at <http://is.gd/dcrhdw>

The manifest problem

Each active line creates a layer

branch: master

[dockerfiles](#) / [meetup](#) / Dockerfile



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0 contributors

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```

Base OS

Sources

Update repos

Install nginx

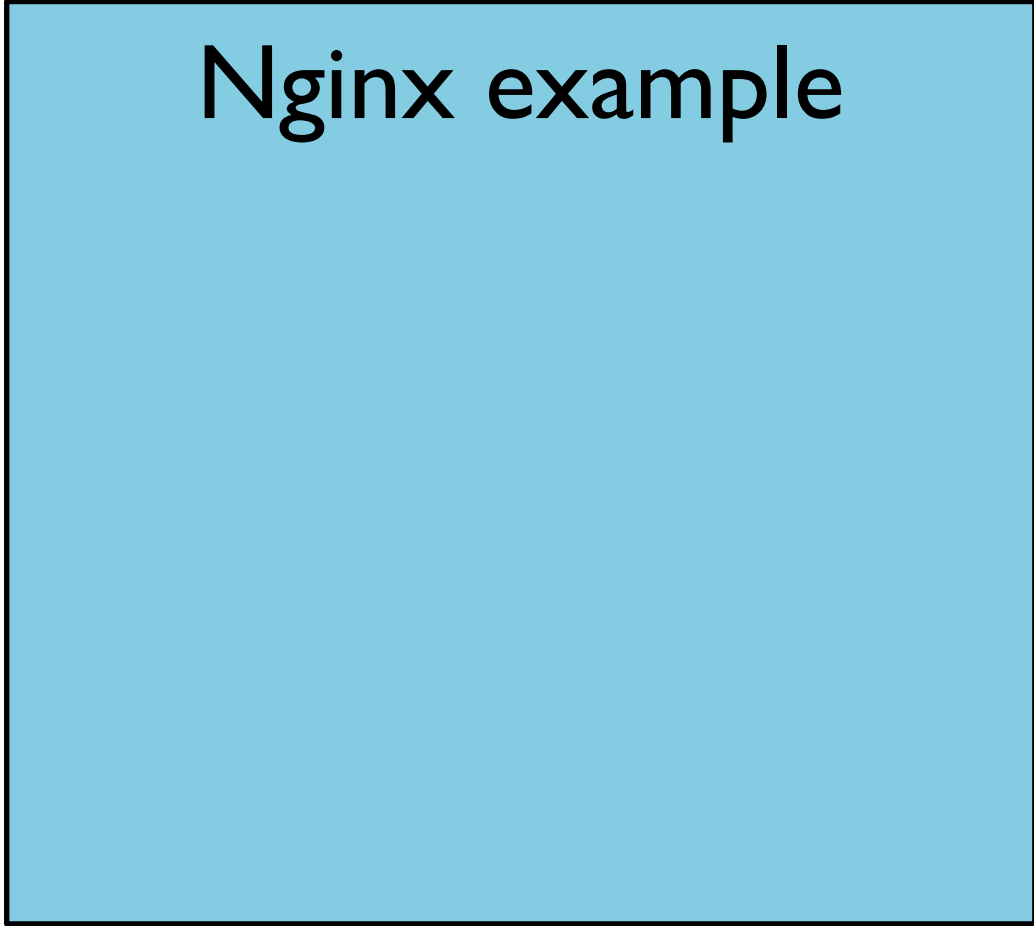
Mod nginx.conf

Mod index.html

An image binds layers together

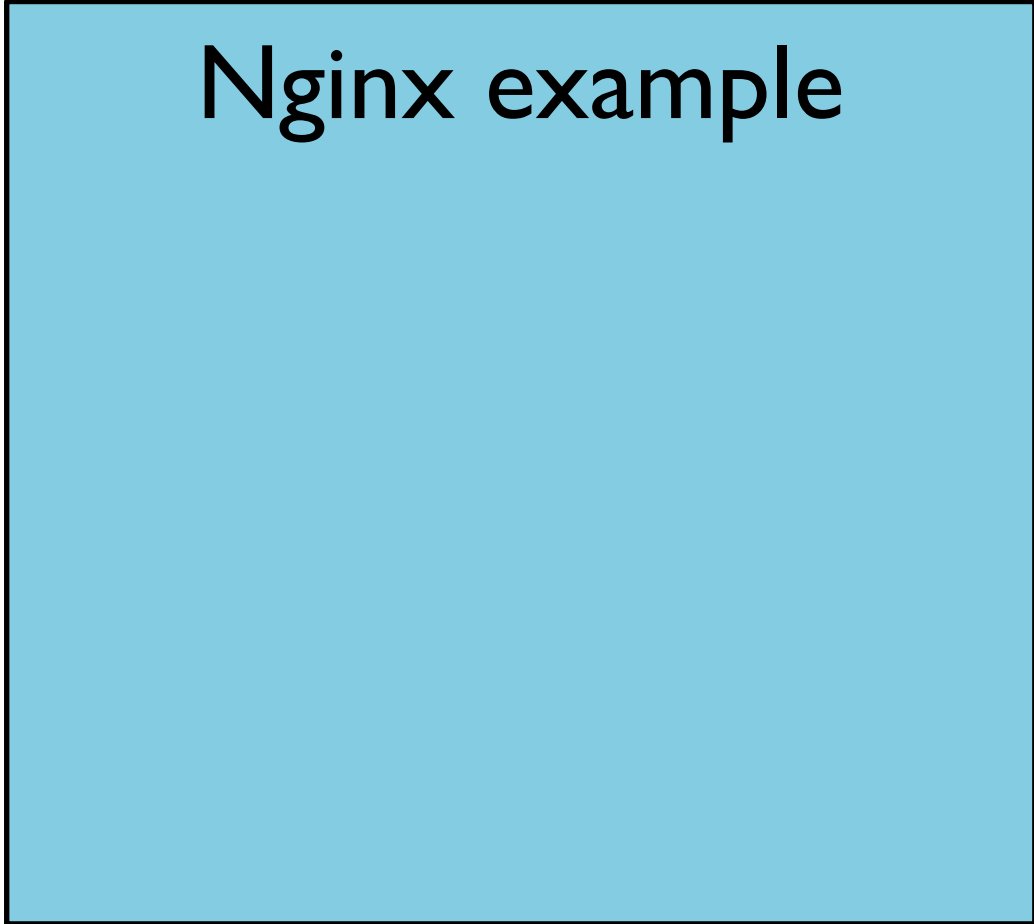


The image is the unit of deployment



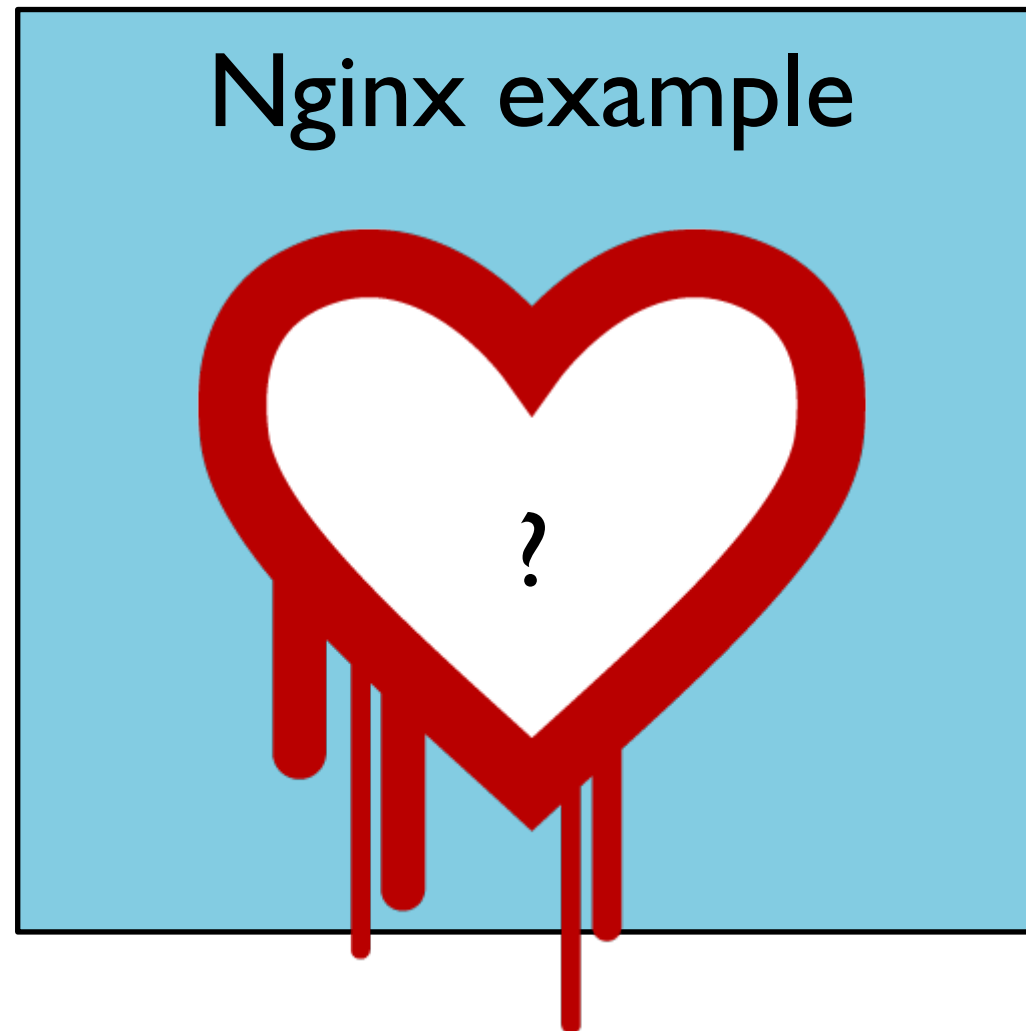
Nginx example

What version of nginx is that?



Nginx example

What version of OpenSSL installed?



and which bash?

Nginx example



My Docker wish list

Feb 2014

Docker CLI

Disk quotas

Route propagation

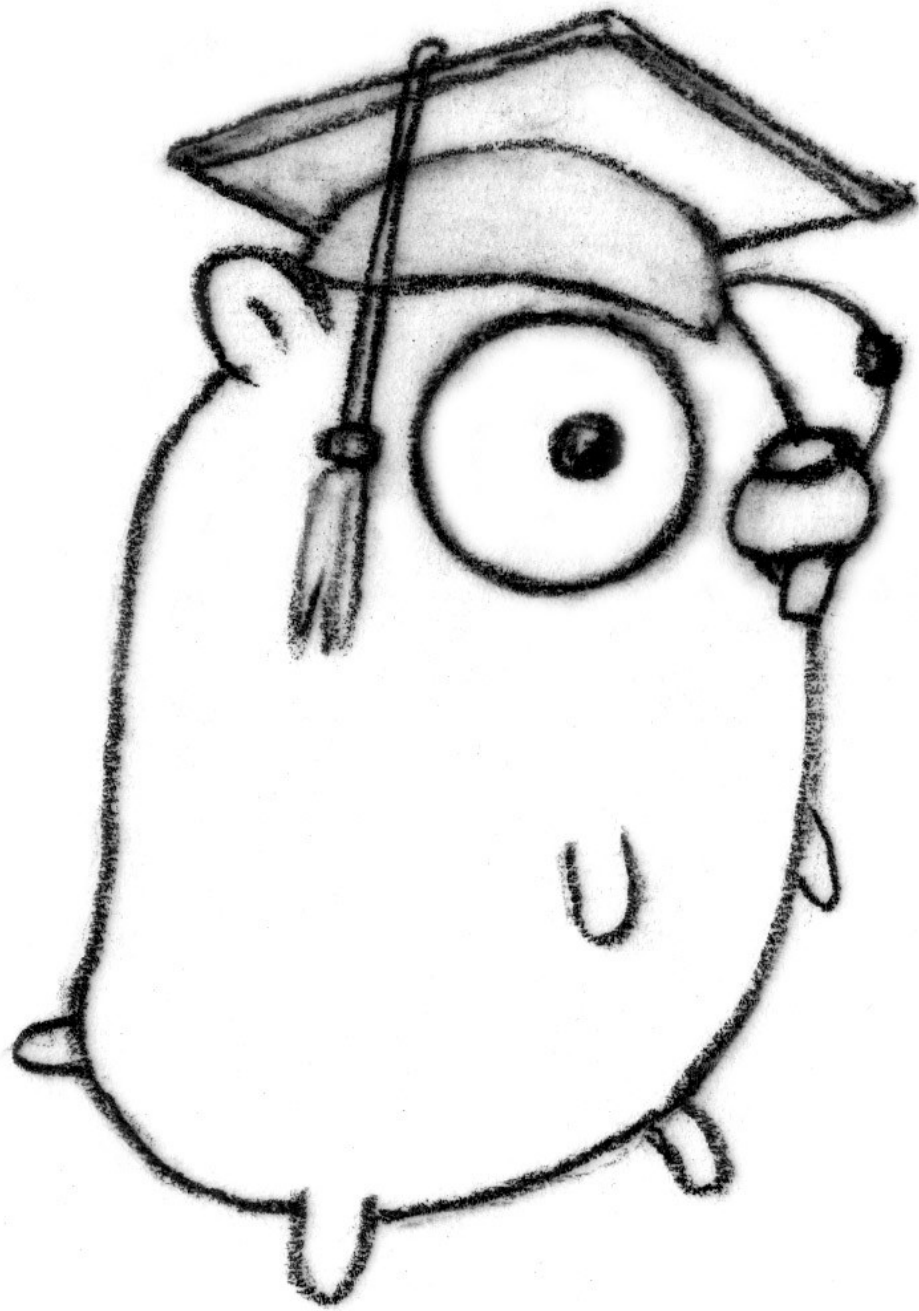
Nov 2014

Docker CLI

Disk quotas

Finer life cycle control

If you want a better Docker



Learn Go

And another thing...

Windows



Governance

1. Mature tooling/CI process so project can scale (more)
2. Dealing with long outstanding pull requests
3. Clear separation between Docker the company and Docker the project
(and clarification of MS relationship)
4. Clear articulation of roadmap

Maintainers



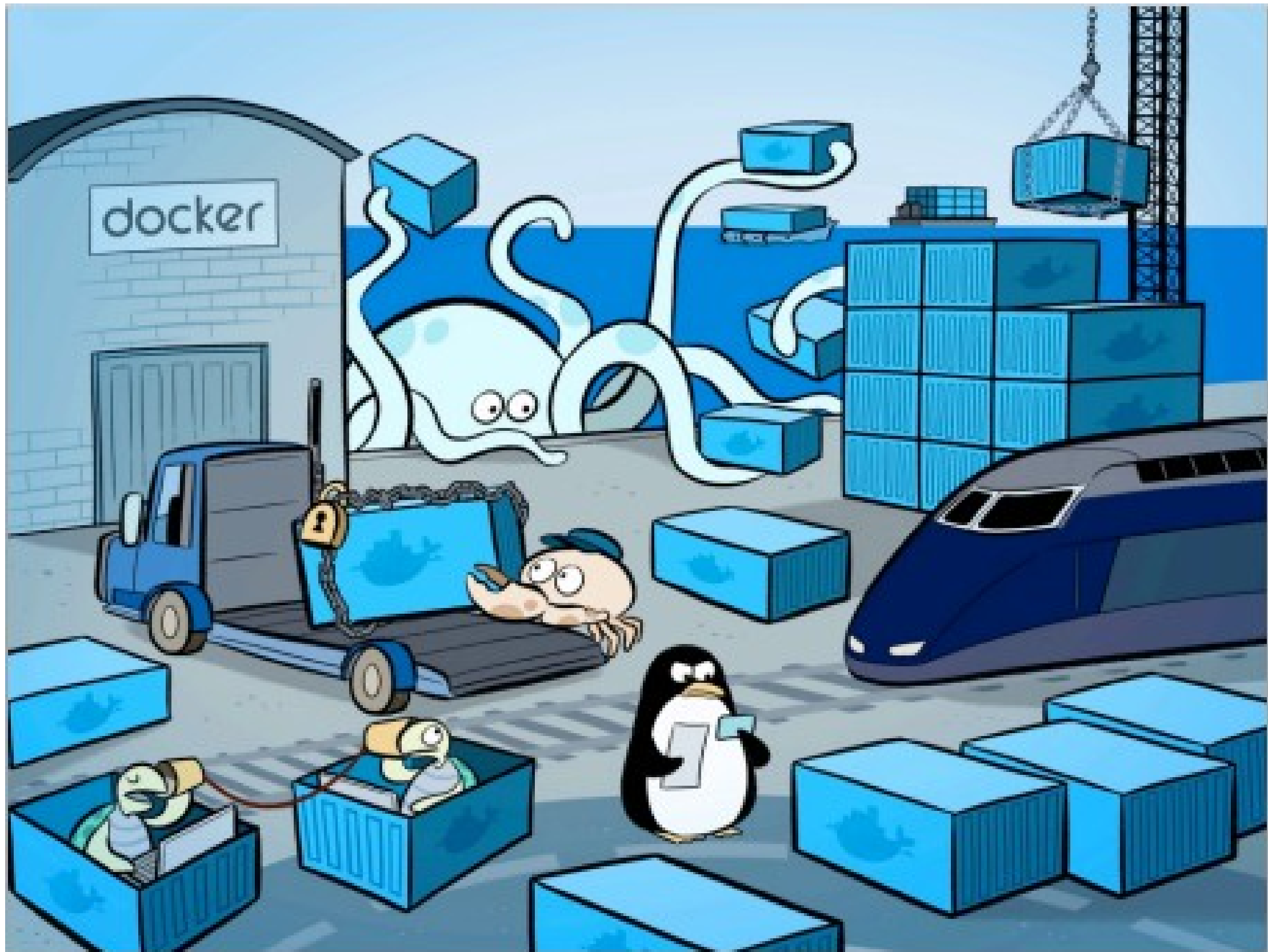
Solomon Hykes

@solomonstre

A maintainer is a contributor of decisions rather than code

How did we come so far so fast?

They used Docker



Wrapping up

TL;DR

Docker is important for DevOps because it tightens iteration loops

The core project developed very fast, but the ecosystem is now moving faster

They did it with Docker

Questions?

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